Appendix II. Debt Sustainability Analysis Update

1. **This debt sustainability analysis (DSA) updates the joint IMF-World Bank DSA** presented to the Boards of both institutions in October 2013 and uses a unified discount rate of 5 percent to calculate the present value of external debt and to assess concessionality. At that time, the DSA results showed that Sierra Leone's risk of debt distress remains moderate.[1] Under the baseline scenario, all debt indicators were below their respective policy-dependent indicative thresholds throughout the projection period (2013–33). The analysis also indicated that the medium- to long-term debt outlook was vulnerable to adverse shocks to several macroeconomic variables, notably exports, commodity prices, foreign direct investment (FDI) inflows, government revenue, and borrowing conditions. The current analysis reaches the same conclusion. Consequently, staff reiterated the need for: continued prudent borrowing policies, sustained fiscal consolidation efforts, continued implementation of growth-enhancing structural reforms, and promotion of economic diversification. The authorities agreed with Staff's assessment and policy recommendations.

2. **The baseline macroeconomic assumptions underlying this DSA update are:**

- **The economy is expected to grow by 8.0 percent in 2014, and by an average rate of about 6.5 percent per year during 2015–19**, driven by public investment scaling up in the context of the authorities' new Poverty Reduction Strategy (PRS), iron ore production and output expansion in agriculture. Over the long-term, growth would moderate to about 5.4 percent.

- **Inflation is projected to rise in the medium term**. Disruption of activity due to the Ebola outbreak is expected to push up inflation from 8.8 percent in 2014 to 10.1 percent in 2015. Inflation will then decline gradually and hover around 5.4 percent in the long term.

- **The overall fiscal deficit is projected to improve over the medium-to-long term**, with the domestic primary deficit expected to improve from 3.9 percent of GDP in 2014 to 2.3 percent in 2019; and to stabilize around 2 percent during 2020–34. Revenue and grants are projected to increase from 13 percent of GDP in 2014 to around 16 percent during the projection period as revenue recovers. Primary (non-interest) expenditure is projected to rise from 16.8 percent of GDP in 2013 to 19 percent over the medium term, on the assumption

[1] This Debt Sustainability Analysis (DSA) is based on the Debt Sustainability Framework (DSF) for low-income countries (LICs), approved by the Executive Boards of the IMF and World Bank. It updates the previous joint DSA of October 21, 2013; and takes into account the decision by the Executive Boards of the IMF and the World Bank to adopt a unified discount rate of 5 percent for the purpose of calculating the present value of the external debt of LICs and assessing the concessionality of individual loans.

that more fiscal resources would be committed to fighting the Ebola Virus Disease and its aftermath.

- **The external current account deficit is forecast to narrow over the medium-to-long-term**, consistent with the projected increase in iron ore and other exports. The projected import dynamics broadly reflects the expected overall real GDP growth.
- **New external loans are projected to average 2.3 percent of GDP over the medium-term** and long term, consistent with the authorities' objectives under the new PRS. The DSA also assumes some nonconcessional borrowing limited at 0.5 percent of GDP (US$30 million) per year during the projection period to finance key priority projects for which the authorities would not be able to secure concessional funding.
- **Domestic debt is projected to rise from 9.2 percent of GDP in 2013, to 10.3 percent by the end of the projection period**, mainly reflecting sustained financing constraints for the budget in the face of persistent needs for increased public investment.

3. **Compared with the previous DSA, the key changes are as follows** (Text Table 1): (i) real GDP growth is expected to be lower than in the 2013 DSA in the short-to-medium term, due to the Ebola outbreak and its lingering effects. Growth is projected to slow from 11 ¼ percent to 8 percent in 2014 due to disruptions in economic activity. Recovery to the long-term growth trajectory is expected to take place only gradually; (ii) a higher primary deficit, particularly in the short run, because of Ebola-related revenue shortfalls and spending needs; (iii) a higher level of domestic borrowing, in line with a weaker fiscal position and; (iv) the external current account deficit is expected to be higher as export growth decelerates, in line with reduced domestic production, while imports increase to cover additional Ebola-related needs.

Text Table 1. Selected Economic Indicators, 2013–34

(Percent of GDP[1], unless otherwise indicated)

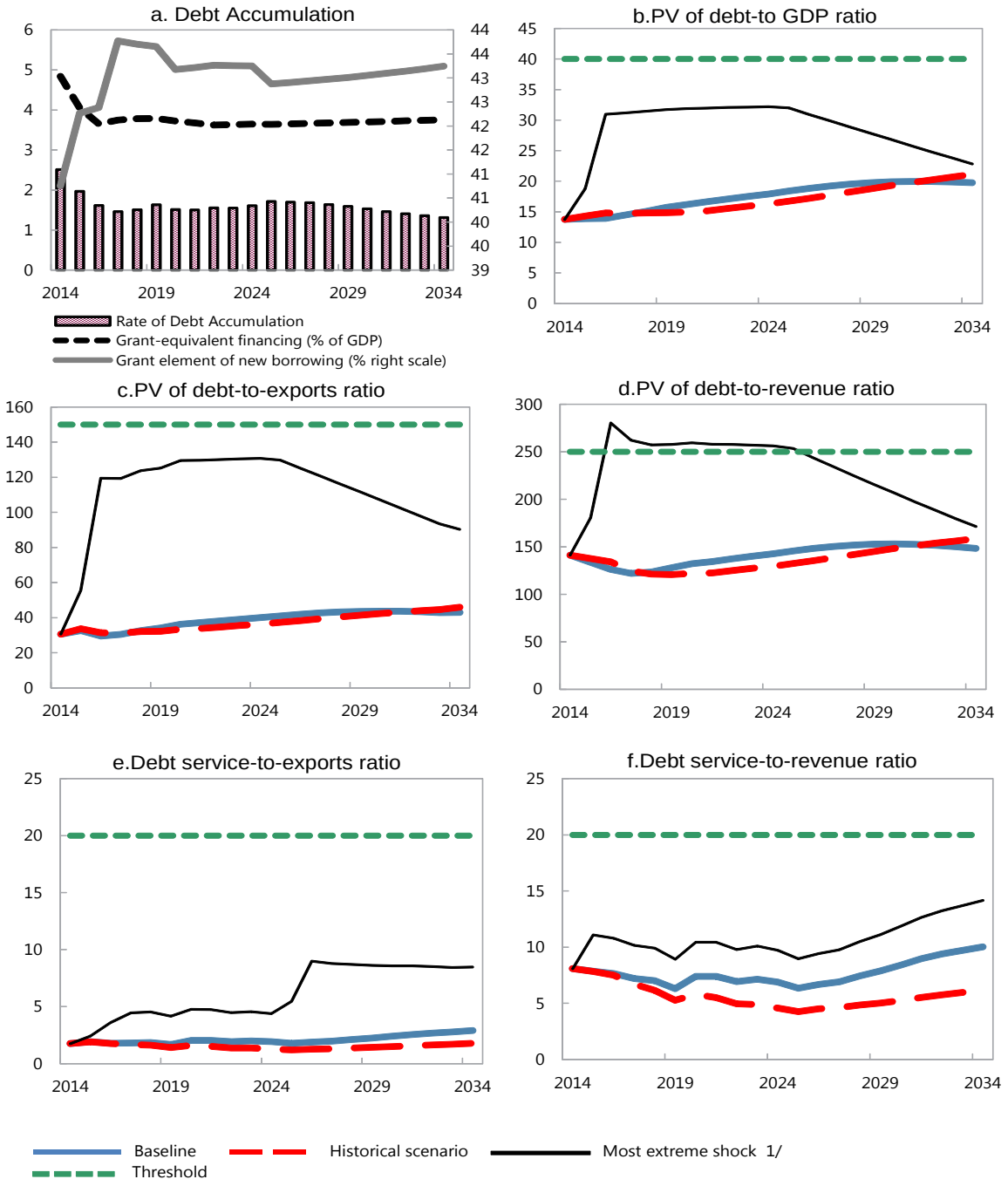
	2013	2014	2015	2016	2017	2018	2019	Long Term[2]
Real GDP Growth (in percent)								
Current DSA	20.1	8.0	10.4	7.8	4.8	4.8	4.9	5.4
Previous DSA	13.3	14.0	12.4	7.7	5.2	5.3	5.0	5.4
Primary fiscal deficit								
Current DSA	1.1	3.6	3.9	2.6	2.6	2.4	2.3	2.0
Previous DSA	1.5	2.6	2.3	2.2	1.8	1.6	1.7	1.6
Current account deficit								
Current DSA	10.4	10.8	8.1	6.6	6.7	6.3	5.9	4.2
Previous DSA	16.4	8.7	6.1	5.8	5.7	4.5	5.7	4.1
Foreign direct investment								
Current DSA	7.3	5.6	5.4	4.5	3.8	3.5	2.9	2.5
Previous DSA	13.4	6.0	3.8	3.5	4.1	3.5	3.4	2.4
External debt								
Current DSA	21.3	23.3	23.2	23.1	23.9	24.6	25.4	28.6
Previous DSA	24.2	23.5	22.8	23.2	23.6	24.1	24.7	27.3
Domestic debt								
Current DSA	9.2	9.8	9.9	10.0	10.5	10.7	10.6	11.9
Previous DSA	11.0	10.3	9.9	9.9	9.9	9.6	9.5	9.6

Sources: The Sierra Leone Authorities and IMF staff projections.

1/ GDP includes iron ore activity.

2/ For the current DSA, the long term covers the period 2020-34, and for the previous DSA it covers the period 2019-33.

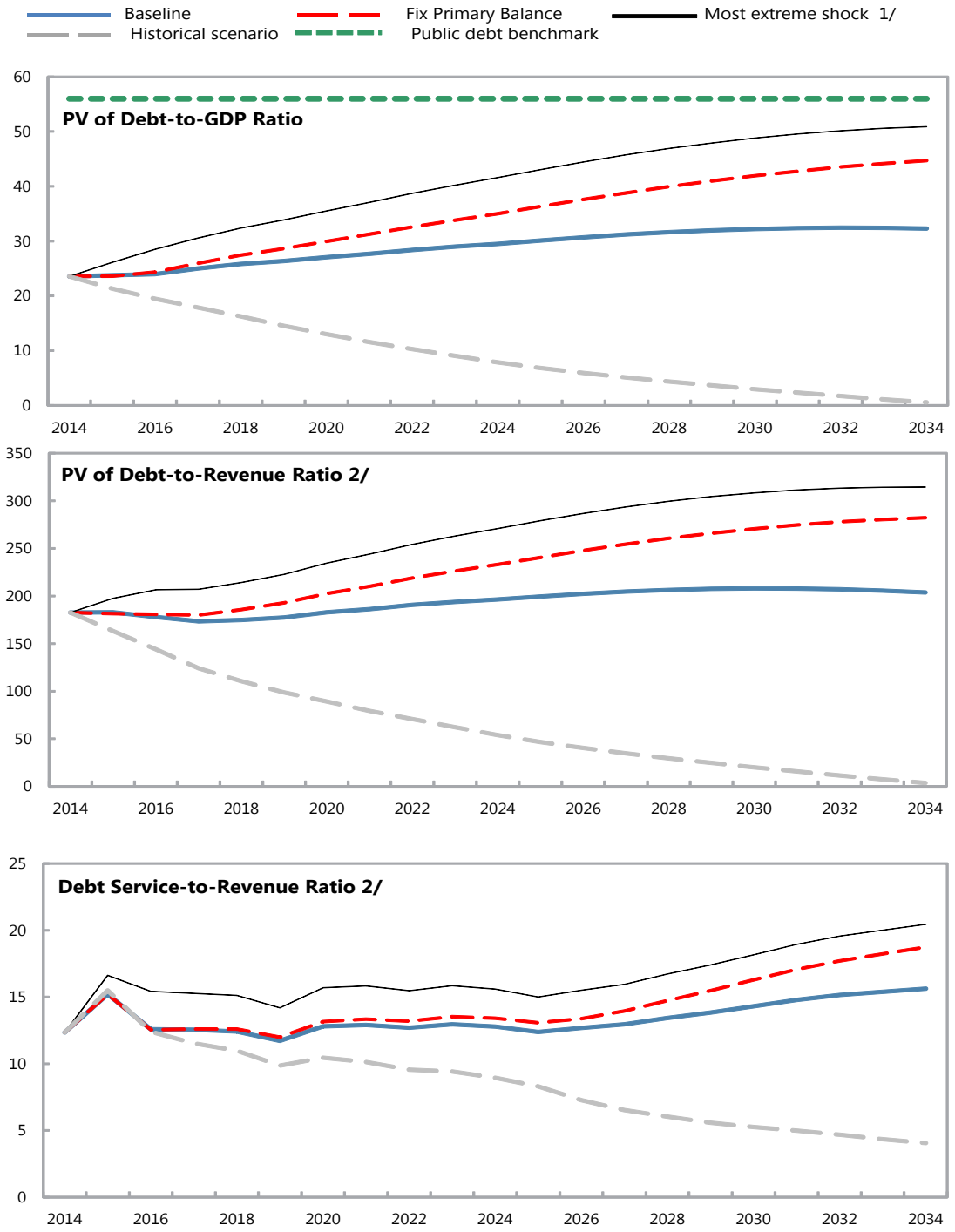
Figure 1. Sierra Leone: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Sierra Leone: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 1. Sierra Leone: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**33.3**	**26.2**	**21.3**			**23.3**	**23.2**	**23.1**	**23.9**	**24.6**	**25.4**		**27.9**	**29.6**	
of which: public and publicly guaranteed (PPG)	33.3	26.2	21.3			23.3	23.2	23.1	23.9	24.6	25.4		27.9	29.6	
Change in external debt	1.2	-7.0	-4.9			1.9	0.0	-0.1	0.8	0.8	0.8		0.4	-0.1	
Identified net debt-creating flows	4.4	4.6	-2.9			3.8	0.7	0.7	2.0	1.9	2.0		1.2	-0.5	
Non-interest current account deficit	**44.7**	**28.6**	**10.4**	**13.2**	**13.9**	**10.8**	**8.1**	**6.6**	**6.7**	**6.3**	**5.9**		**4.9**	**2.9**	4.2
Deficit in balance of goods and services	52.1	30.2	3.2			3.9	4.4	-0.7	-2.3	-1.0	-1.3		-0.7	-5.3	
Exports	18.6	35.6	43.9			45.0	42.5	47.2	47.6	46.3	46.1		44.9	46.0	
Imports	70.7	65.8	47.0			48.9	46.9	46.5	45.3	45.2	44.8		44.1	40.7	
Net current transfers (negative = inflow)	-8.5	-6.2	-4.1	-6.4	1.8	-4.5	-3.9	-4.0	-4.0	-4.0	-3.9		-3.9	-4.1	-4.0
of which: official	-3.8	-2.5	-1.0			-1.5	-1.0	-1.1	-1.1	-1.1	-1.1		-1.2	-1.6	
Other current account flows (negative = net inflow)	1.0	4.6	11.3			11.4	7.7	11.3	13.0	11.3	11.1		9.6	12.3	
Net FDI (negative = inflow)	**-36.6**	**-16.7**	**-7.3**	**-9.7**	**10.6**	**-5.6**	**-5.4**	**-4.5**	**-3.8**	**-3.5**	**-2.9**		**-2.7**	**-2.1**	-2.5
Endogenous debt dynamics 2/	**-3.7**	**-7.3**	**-6.0**			**-1.4**	**-2.0**	**-1.5**	**-0.9**	**-0.9**	**-1.0**		**-1.1**	**-1.3**	
Contribution from nominal interest rate	0.2	0.2	0.0			0.1	0.1	0.1	0.1	0.2	0.2		0.3	0.3	
Contribution from real GDP growth	-1.7	-3.9	-4.1			-1.5	-2.1	-1.6	-1.0	-1.1	-1.1		-1.4	-1.6	
Contribution from price and exchange rate changes	-2.2	-3.6	-1.9			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.1**	**-11.6**	**-2.0**			**-1.9**	**-0.7**	**-0.8**	**-1.2**	**-1.2**	**-1.2**		**-0.7**	**0.3**	
of which: exceptional financing	0.0	0.0	0.0			-1.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	12.4			13.8	13.9	13.9	14.5	15.1	15.7		17.9	19.8	
In percent of exports	...	...	28.3			30.6	32.6	29.6	30.5	32.6	34.1		40.0	43.0	
PV of PPG external debt	**...**	**...**	**12.4**			**13.8**	**13.9**	**13.9**	**14.5**	**15.1**	**15.7**		**17.9**	**19.8**	
In percent of exports	**...**	**...**	**28.3**			**30.6**	**32.6**	**29.6**	**30.5**	**32.6**	**34.1**		**40.0**	**43.0**	
In percent of government revenues	**...**	**...**	**116.2**			**141.1**	**133.5**	**126.2**	**122.0**	**123.5**	**127.9**		**142.6**	**148.3**	
Debt service-to-exports ratio (in percent)	**3.3**	**1.6**	**0.0**			**1.8**	**1.9**	**1.8**	**1.8**	**1.9**	**1.7**		**1.9**	**2.9**	
PPG debt service-to-exports ratio (in percent)	**3.3**	**1.6**	**0.0**			**1.8**	**1.9**	**1.8**	**1.8**	**1.9**	**1.7**		**1.9**	**2.9**	
PPG debt service-to-revenue ratio (in percent)	**5.3**	**5.0**	**0.0**			**8.1**	**7.9**	**7.7**	**7.2**	**7.0**	**6.3**		**6.9**	**10.0**	
Total gross financing need (Billions of U.S. dollars)	0.3	0.5	0.2			0.3	0.2	0.2	0.3	0.3	0.3		0.4	0.5	
Non-interest current account deficit that stabilizes debt ratio	43.4	35.7	15.3			8.9	8.2	6.7	5.9	5.5	5.1		4.5	3.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.0	15.2	20.1	7.9	5.5	8.0	10.4	7.8	4.8	4.8	4.9	6.8	5.3	5.8	5.4
GDP deflator in US dollar terms (change in percent)	7.4	12.1	8.0	5.6	6.0	1.9	2.8	3.1	1.4	1.4	1.5	2.0	1.4	1.5	1.4
Effective interest rate (percent) 5/	0.7	0.7	0.0	0.8	0.5	0.6	0.7	0.7	0.6	0.7	0.7	0.7	1.0	1.2	1.0
Growth of exports of G&S (US dollar terms, in percent)	28.9	146.8	60.1	30.3	45.0	12.8	7.3	23.3	7.2	3.3	6.2	10.0	6.9	6.4	6.9
Growth of imports of G&S (US dollar terms, in percent)	90.1	20.1	-7.2	23.4	37.3	14.4	8.9	10.1	3.5	6.3	5.5	8.1	5.5	6.5	6.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	40.8	42.3	42.4	43.8	43.7	43.7	42.8	43.2	43.2	43.1
Government revenues (excluding grants, in percent of GDP)	11.5	11.4	10.7			9.8	10.4	11.1	11.9	12.2	12.3		12.6	13.3	12.8
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.2			0.3	0.3	0.3	0.3	0.4	0.4		0.5	1.1	
of which: Grants	0.2	0.1	0.1			0.2	0.2	0.2	0.2	0.2	0.2		0.3	0.6	
of which: Concessional loans	0.1	0.1	0.1			0.1	0.1	0.1	0.2	0.2	0.2		0.3	0.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.8	4.1	3.7	3.7	3.8	3.8		3.6	3.8	3.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			66.0	67.5	68.5	70.3	70.2	70.2		70.0	69.5	69.8
Memorandum items:															
Nominal GDP (Billions of US dollars)	2.9	3.8	4.9			5.4	6.1	6.8	7.2	7.7	8.2		11.3	22.3	
Nominal dollar GDP growth	13.8	29.1	29.8			10.1	13.5	11.1	6.2	6.3	6.5	8.9	6.7	7.4	6.9
PV of PPG external debt (in Billions of US dollars)			0.6			0.7	0.8	0.9	1.0	1.1	1.3		2.0	4.4	
(PVt-PVt-1)/GDPt-1 (in percent)						2.5	2.0	1.6	1.5	1.5	1.6	1.8	1.6	1.3	1.5
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.1	0.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	12.3			13.7	13.8	13.9	14.4	15.0	15.6		17.8	19.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	27.8			30.1	32.1	29.2	30.1	32.2	33.7		39.6	42.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.0			1.7	1.9	1.8	1.8	1.8	1.7		1.9	2.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	14	14	14	15	15	16	**18**	20
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	14	14	15	15	15	15	**16**	21
A2. New public sector loans on less favorable terms in 2014-2034 2/	14	14	15	16	18	19	**24**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	14	15	16	16	17	18	**20**	22
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	14	19	31	31	31	32	**32**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	14	14	15	15	16	17	**19**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	14	17	20	20	21	21	**23**	21
B5. Combination of B1-B4 using one-half standard deviation shocks	14	14	18	19	19	20	**22**	22
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	14	19	19	20	21	22	**25**	27
PV of debt-to-exports ratio								
Baseline	31	33	30	31	33	34	**40**	43
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	31	34	31	31	32	32	**36**	46
A2. New public sector loans on less favorable terms in 2014-2034 2/	31	34	32	34	38	41	**53**	65
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	31	32	29	30	32	34	**39**	42
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	31	56	120	119	124	125	**131**	90
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	31	32	29	30	32	34	**39**	42
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	31	40	42	42	44	46	**51**	45
B5. Combination of B1-B4 using one-half standard deviation shocks	31	32	40	41	43	45	**51**	50
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	31	32	29	30	32	34	**39**	42
PV of debt-to-revenue ratio								
Baseline	141	134	126	122	123	128	**143**	148
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	141	138	134	124	121	121	**129**	158
A2. New public sector loans on less favorable terms in 2014-2034 2/	141	139	137	138	144	153	**190**	224
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	141	142	141	136	138	143	**159**	166
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	141	181	280	262	257	258	**256**	171
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	141	136	133	128	130	135	**150**	156
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	141	164	178	169	168	171	**180**	154
B5. Combination of B1-B4 using one-half standard deviation shocks	141	137	163	156	156	160	**173**	163
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	141	186	176	170	172	178	**198**	206

Table 2. Sierra Leone: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	2	2	2	2	2	2	**2**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	2	2	2	2	2	1	**1**	2
A2. New public sector loans on less favorable terms in 2014-2034 2/	2	2	2	2	2	2	**3**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	2	2	2	2	2	2	**2**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	2	2	4	4	5	4	**4**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	2	2	2	2	2	2	**2**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	2	2	2	2	2	2	**2**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	2	2	**2**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	2	2	2	2	2	2	**2**	3
Debt service-to-revenue ratio								
Baseline	8	8	8	7	7	6	**7**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	8	8	8	7	6	5	**5**	6
A2. New public sector loans on less favorable terms in 2014-2034 2/	8	8	8	8	8	8	**10**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	8	8	9	8	8	7	**8**	11
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	8	8	8	10	9	9	**9**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	8	8	8	8	8	7	**7**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	8	8	8	8	8	7	**7**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	8	8	8	7	**8**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	8	11	11	10	10	9	**10**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	41	41	41	41	41	41	**41**	41

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Sierra Leone: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	44.9	36.9	30.5			33.0	33.1	33.1	34.4	35.4	36.0		39.5	42.1	
of which: foreign-currency denominated	33.5	26.3	21.3			23.3	23.2	23.1	23.9	24.6	25.4		27.9	29.6	
Change in public sector debt	-1.9	-8.0	-6.4			2.5	0.1	0.0	1.2	1.0	0.7		0.5	-0.1	
Identified debt-creating flows	-4.3	-6.0	-5.2			1.6	0.9	-0.1	1.5	1.2	1.0		0.8	-0.2	
Primary deficit	2.6	3.5	1.1	-1.3	7.5	3.6	3.9	2.6	2.6	2.4	2.3	2.9	2.2	1.6	2.0
Revenue and grants	17.0	15.2	13.2			12.9	13.0	13.5	14.4	14.8	14.9		15.0	15.8	
of which: grants	5.6	3.8	2.6			3.1	2.6	2.4	2.5	2.5	2.5		2.5	2.5	
Primary (noninterest) expenditure	19.6	18.7	14.4			16.5	16.9	16.1	17.0	17.2	17.2		17.2	17.5	
Automatic debt dynamics	-6.1	-8.6	-6.2			-1.7	-2.9	-2.6	-1.1	-1.2	-1.3		-1.5	-1.8	
Contribution from interest rate/growth differential	-3.3	-5.8	-5.0			-2.0	-2.7	-2.3	-1.2	-1.3	-1.4		-1.1	-1.4	
of which: contribution from average real interest rate	-0.6	0.2	1.2			0.3	0.4	0.1	0.3	0.3	0.2		0.9	0.9	
of which: contribution from real GDP growth	-2.6	-5.9	-6.2			-2.3	-3.1	-2.4	-1.5	-1.6	-1.7		-1.9	-2.3	
Contribution from real exchange rate depreciation	-2.8	-2.9	-1.3			0.2	-0.2	-0.3	0.1	0.1	0.1		...	...	
Other identified debt-creating flows	-0.7	-0.8	-0.1			-0.2	-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.3	-0.1	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.5	-0.8	0.0			-0.2	-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.3	-2.0	-1.2			0.9	-0.8	0.1	-0.2	-0.2	-0.4		-0.2	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	21.6			23.5	23.8	24.0	25.0	25.8	26.3		29.5	32.3	
of which: foreign-currency denominated	...	...	12.4			13.8	13.9	13.9	14.5	15.1	15.7		17.9	19.8	
of which: external	...	...	12.4			13.8	13.9	13.9	14.5	15.1	15.7		17.9	19.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.0	5.6	2.4			5.2	5.9	4.3	4.4	4.3	4.1		4.1	4.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	163.1			182.8	182.8	177.9	173.4	174.8	177.4		196.4	203.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	202.2			241.4	229.0	216.8	210.0	211.1	214.0		234.8	242.0	
of which: external 3/	...	...	116.2			141.1	133.5	126.2	122.0	123.5	127.9		142.6	148.3	
Debt service-to-revenue and grants ratio (in percent) 4/	14.0	13.9	9.4			12.3	15.2	12.6	12.6	12.4	11.7		12.8	15.6	
Debt service-to-revenue ratio (in percent) 4/	20.8	18.6	11.7			16.3	19.0	15.3	15.2	15.0	14.2		15.3	18.6	
Primary deficit that stabilizes the debt-to-GDP ratio	4.5	11.5	7.5			1.1	3.8	2.6	1.4	1.4	1.7		1.7	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.0	15.2	20.1	7.9	5.5	8.0	10.4	7.8	4.8	4.8	4.9	6.8	5.3	5.8	5.4
Average nominal interest rate on forex debt (in percent)	0.7	0.7	0.0	0.9	0.6	0.6	0.7	0.7	0.6	0.7	0.7	0.7	1.0	1.2	1.0
Average real interest rate on domestic debt (in percent)	-1.9	4.8	6.6	-0.7	4.2	5.5	7.2	3.3	5.7	5.5	5.1	5.4	5.4	5.3	5.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.4	-10.0	-5.6	-3.6	8.5	1.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	17.4	12.0	8.1	12.1	4.0	4.1	6.3	6.3	4.5	4.4	4.5	5.0	4.4	4.5	4.5
Growth of real primary spending (deflated by GDP deflator, in percent)	11.1	9.8	-7.8	1.4	5.4	24.0	13.5	2.2	10.9	6.0	5.0	10.3	5.3	5.9	5.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	40.8	42.3	42.4	43.8	43.7	43.7	42.8	43.2	43.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Sierra Leone: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	24	24	24	25	26	26	30	32
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	21	19	18	16	15	8	1
A2. Primary balance is unchanged from 2014	24	24	24	26	27	29	35	45
A3. Permanently lower GDP growth 1/	24	24	25	26	28	29	37	57
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	24	26	29	31	32	34	42	51
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	24	25	27	28	29	29	32	34
B3. Combination of B1-B2 using one half standard deviation shocks	24	24	25	26	28	29	35	42
B4. One-time 30 percent real depreciation in 2015	24	29	28	28	29	29	31	33
B5. 10 percent of GDP increase in other debt-creating flows in 2015	24	29	29	30	31	31	34	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	183	183	178	173	175	177	196	204
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	183	163	144	124	111	99	54	4
A2. Primary balance is unchanged from 2014	183	181	181	180	186	193	233	282
A3. Permanently lower GDP growth 1/	183	185	183	181	186	193	244	348
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	183	198	207	207	214	223	271	315
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	183	193	202	195	196	198	215	214
B3. Combination of B1-B2 using one half standard deviation shocks	183	183	181	180	186	192	229	262
B4. One-time 30 percent real depreciation in 2015	183	221	208	198	195	195	205	208
B5. 10 percent of GDP increase in other debt-creating flows in 2015	183	226	217	209	209	211	226	219
Debt Service-to-Revenue Ratio 2/								
Baseline	12	15	13	13	12	12	13	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	15	12	12	11	10	9	4
A2. Primary balance is unchanged from 2014	12	15	13	13	13	12	13	19
A3. Permanently lower GDP growth 1/	12	15	13	13	13	12	15	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	12	16	14	14	14	13	15	21
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	12	15	13	13	13	12	13	17
B3. Combination of B1-B2 using one half standard deviation shocks	12	16	13	13	13	13	14	18
B4. One-time 30 percent real depreciation in 2015	12	17	15	15	15	14	16	20
B5. 10 percent of GDP increase in other debt-creating flows in 2015	12	15	13	14	13	12	13	17

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.